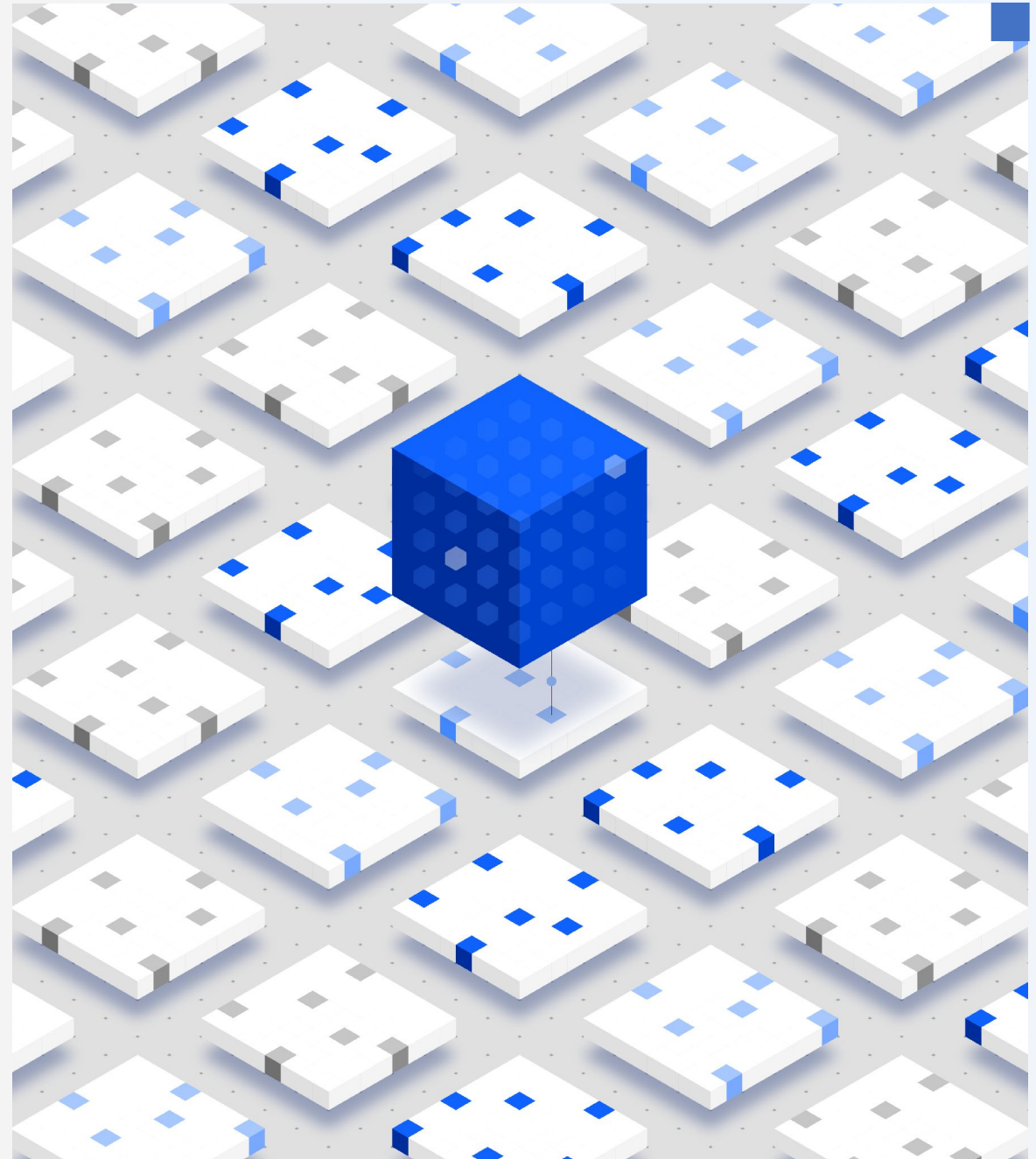


Shift-Left for Data Pipelines

Stop Debugging Pipelines in Production

Sean Anderson
Program Director, watsonx.data
IBM

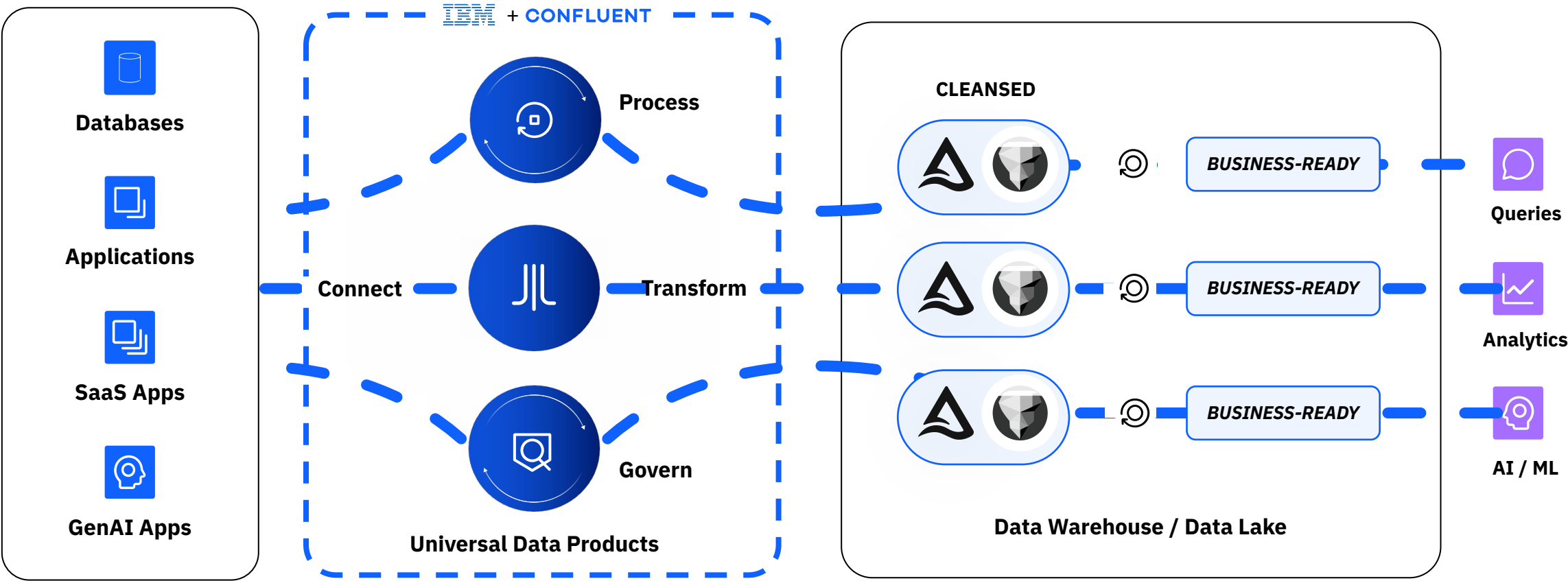


A new approach: shift left to manage data like a product

By moving quality, context, and ownership upstream, IBM enables domain teams to publish universal data products that applications, analytics, and AI agents can reuse.

OPERATIONAL ESTATE

ANALYTICAL ESTATE



Shift-Left Quality Govern closer to source with Schema Registry, Data Quality Rules, and Data Contracts

Companies must be ready to "shift left" and do the work before the data *lands*.

Integration

Process and transform data *before* it lands in your platform—saving on processing costs.

Governance

Contained at the source: schema registry, data contracts, and lineage attached from the first hop so trust travels with the data.

Quality

Producers apply checks upstream: contracts and quality rules validate, cleanse, standardize, and match before data propagates.

How IBM enables shift-left

IBM allows data integration and transformation to happen close to your sources—in your VPC, your data center, so that shift-left is an estate-wide operating principle. The output is a reusable, discoverable data product—consumable as a stream or a table—so you can build data once, reuse it anywhere. [Same work, but upstream rates](#) (and your data platforms stay put).

A Modern Tale

Of Pipeline Calamity

Modern tale of pipeline calamity.

MAR ○

A pipeline ships.

It passes review. Nobody reads it twice.

```
orders_ingest_dag.py
```

60 lines · @daily · works

JUN ○

It multiplies.

One pattern. Three dialects. Zero shared code.

```
orders_ingest_dag.py
```

the original

```
customers_ingest_dag.py
```

copied · renamed · re-scheduled

```
payments_ingest_dag.py
```

copied · a try/except added

OCT ○

The business wants real-time.

The batch team writes streaming the only way it knows how.

```
orders.events.v1
```

Confluent Cloud topic

```
kafka_orders_stream_dag.py
```

```
while True: consumer.poll(1.0)
scheduled every 5 minutes
```

DEC ○

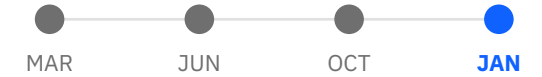
2:04 a.m. Everything is green. Nothing is loading.

Airflow said success. The data said nothing at all.

```
payments_ingest      SUCCESS
rows loaded           0
consecutive days      9
```

The postmortem

Every finding, and the month it actually shipped



Swallowed exception

JUN

the one that paged us

Parse-time DB connections

MAR

run by the scheduler every 30 seconds

Credentials in the code

MAR

multiplied by every copy

datetime.now() windows

MAR

reruns produce different data

Copy-paste drift

JUN

one bug, three files, zero fixes

Blocking consumer loop

OCT

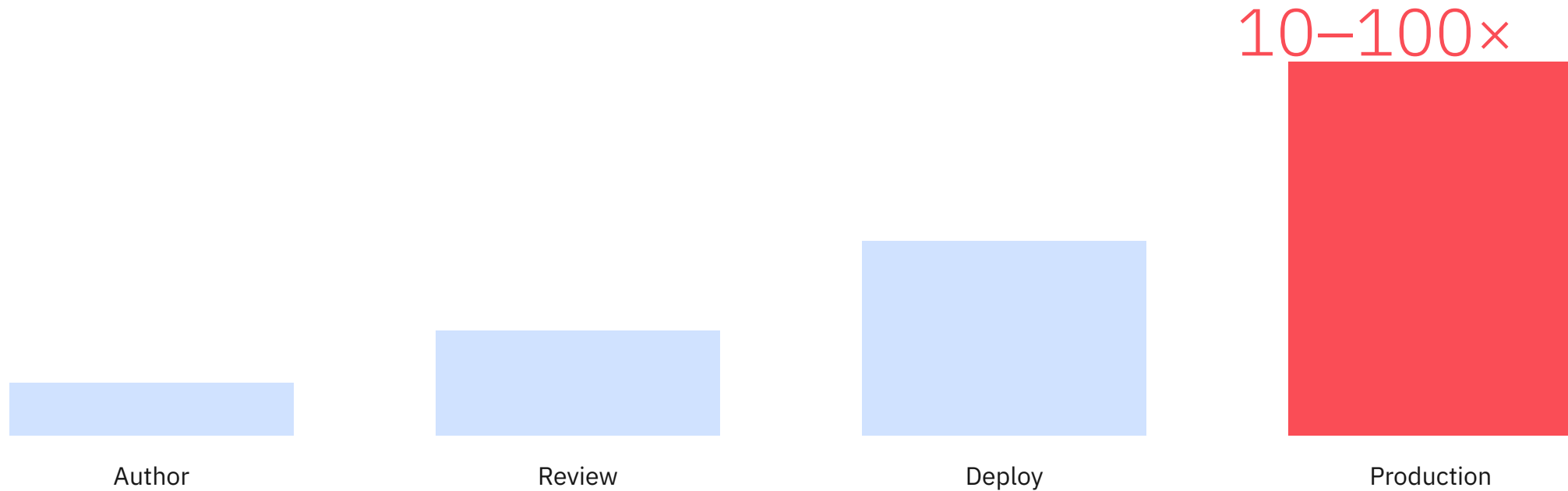
a worker slot, held hostage

None of this needed production to be found.

Rewind.

What if the findings had arrived with the code?

cost of the same fix



Shift-left: move the finding to where the fixing is cheap.

Why this is finally possible

The reviewer moved into the IDE

85%

say AI moved the bottleneck from writing code to trusting it

DevSecOps professionals — IBM, July 2026

IBM Bob

an AI agent that reviews as you write — IDE and terminal

Multi-agent

subagents read whole estates, not open files

Runtime-aware

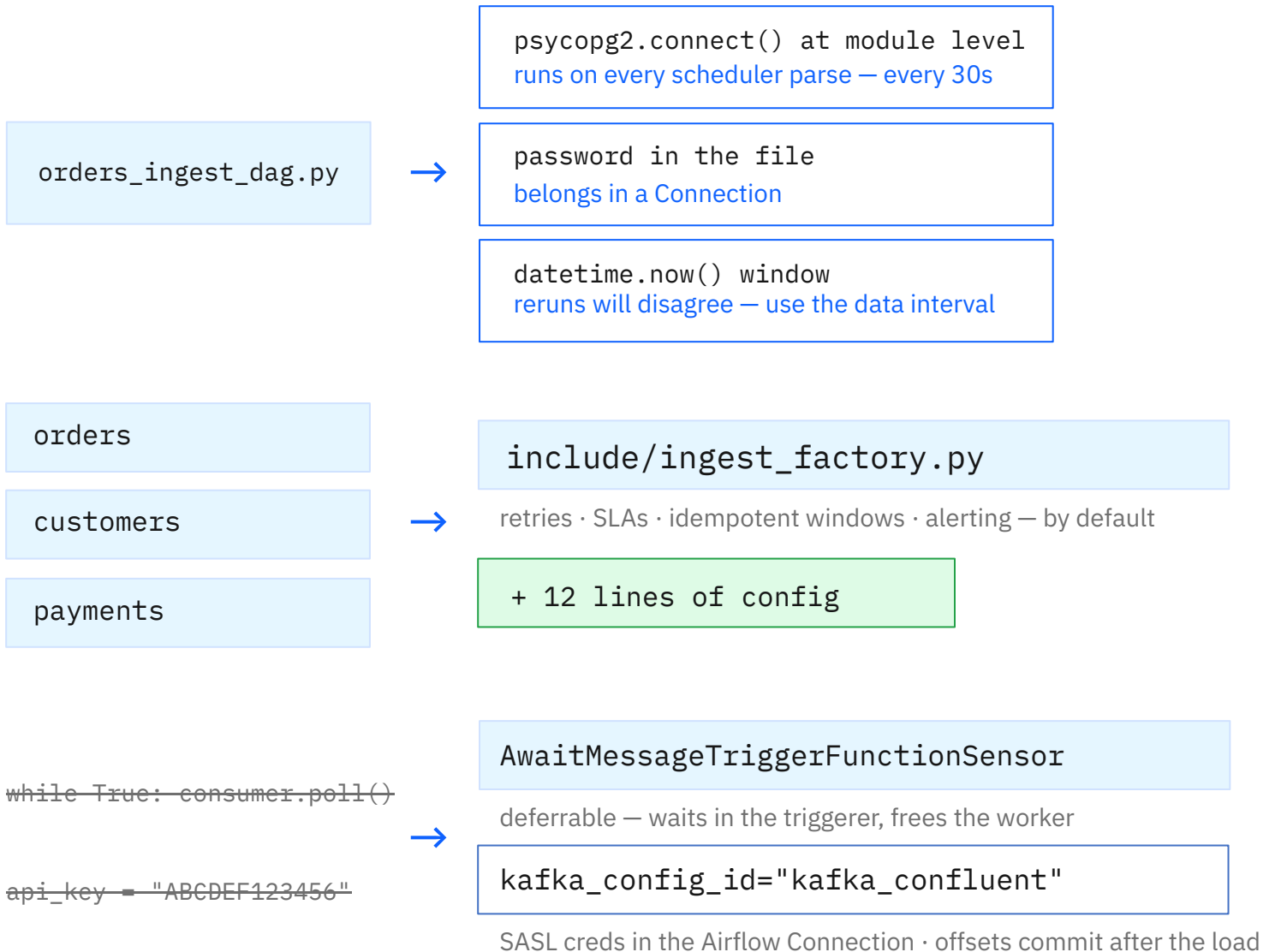
reasons like the scheduler, not like a linter

The replay...

MAR ○ Catch issues before first commit.

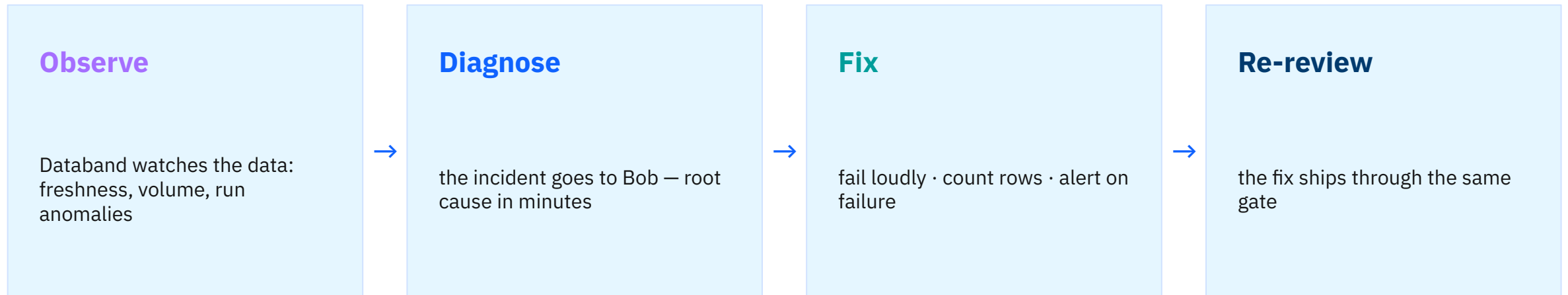
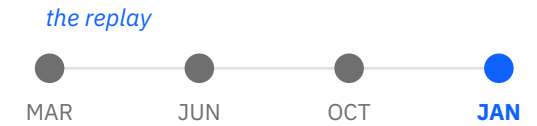
JUN ○ Turns copies into a reusable pattern.

OCT ○ Apply one standard across batch and streaming.



January still comes.

Schema changes, source outages, silent drift — review can't see the future



What review can't catch, observability must — and it feeds the next review.

Epilogue: agents on both sides of the pipeline

On the code

Bob — review, refactor, debug, in the IDE and terminal. Today's story.

On the pipelines

Agentic data integration in watsonx.data integration — agents that author and operate governed flows, with observability built in.

Reliability isn't inspected in after a failure. It's designed in from March.

Watsonx.data + Confluent

Empowering a Shift-left Value Center

Flexible pipeline authoring →

Build pipelines with natural language, drag-and-drop, or SDK-based authoring, helping users of all skill levels get trusted data faster.



Design the pipeline

Assemble sources, transformations and targets — batch, streaming or document — and connect them end to end.



Connect to your data

Register connections to databases, files, cloud storage and message streams. Credentials are validated before anything is saved.



Run on a schedule

Turn any pipeline into a job — start it, stop it, and track every run from your own scheduler or release process.



See what happened

Live throughput, error counts and per-step timings while a run is still in flight, plus the logs behind them.



Keep control

Projects, roles and access groups work exactly as they do in the UI. Generated code hides credentials by default.



Turn what exists into code

Helps identify dependent tasks and datasets when a data quality error occurs so companies have full transparency and can prioritize remediation.

20 years of proven innovation and leadership in data integration

Leader in the 2025 Gartner Magic Quadrant for Data Integration Tools for the 20th consecutive year.

Leader in the 2025 IDC MarketScape for Worldwide Data Integration Software Platforms

Leader in the 2025 Gartner® Magic Quadrant™ for Metadata Management Solutions

Leader in the 2024 Gartner Magic Quadrant for Data and Analytics Governance Platforms

Figure 1: Magic Quadrant for Data Integration Tools



Secure and Governed

Run trusted AI at scale

IBM Context Layer

Define and deliver trusted context for AI

 **Context Hub** | Knowledge Graph • Skills Repository • Context Runtime

 **OpenRAG** | Agentic RAG

 **watsonx.data intelligence** | Lineage • Unified Catalog

 **Real-time Context Engine**

Ecosystem Interoperability



IBM Composable Data Layer

Intelligent Data Curation for Real-Time Context

IBM Data Lakehouse for Analytics

Fit-for-purpose engines

Match every workload to the right engine



Cache

Query Acceleration



Open format storage

Keep data open, portable, reusable



IBM Operational Data



IBM Streaming Data



CONFLUENT

IBM Agentic Data Engineering

Use agents to process and prepare data for intelligent systems



Docling



Tableflow



StreamSets



DataStage



Data Replication



Data Observability

Hybrid Foundation



IBM Cloud

Secure + Governed

Run trusted AI at scale

Security and Privacy

Data Governance and Lineage

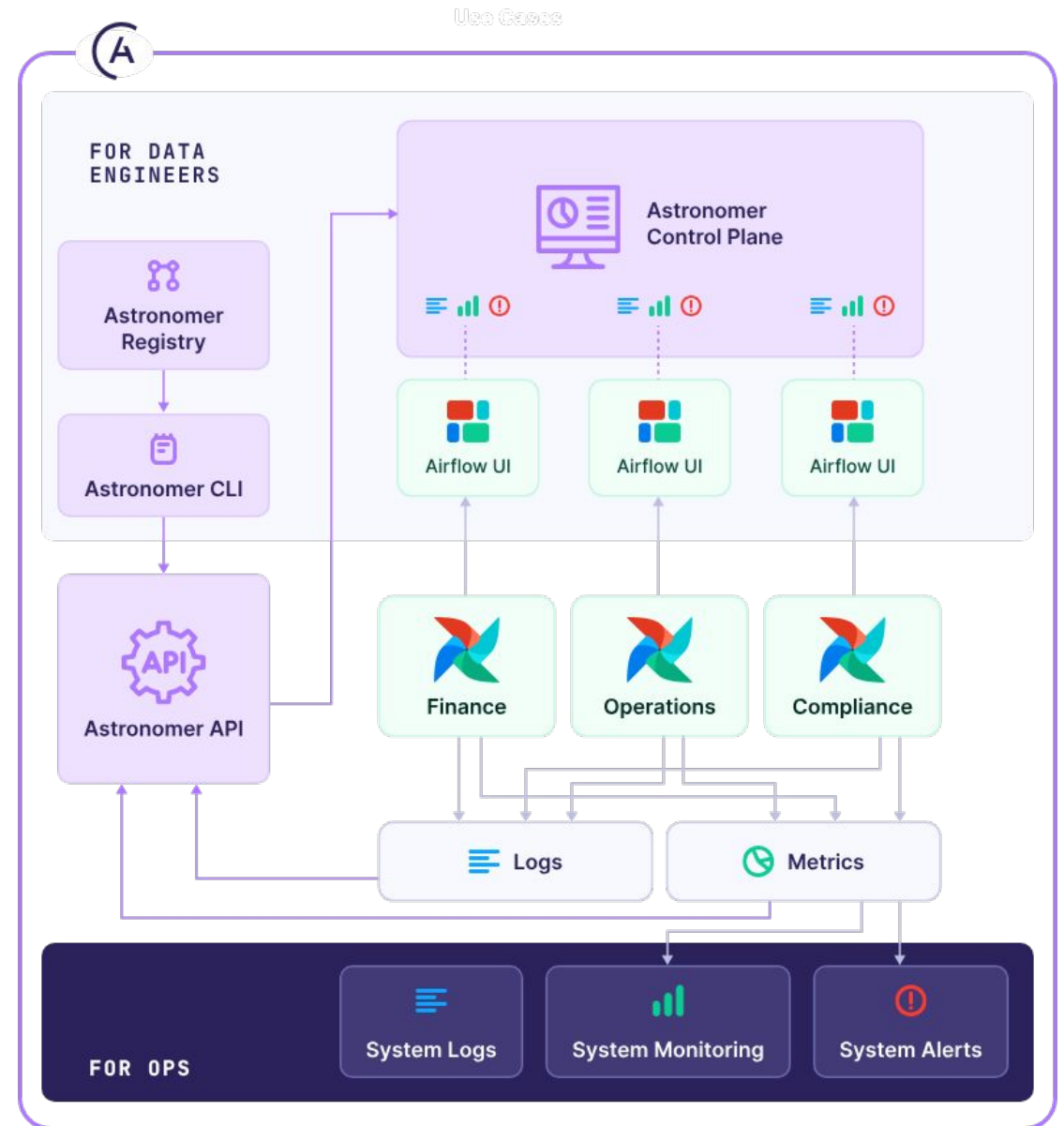
Compliance and Policy

Observability and Quality

FinOps and consumption pricing

Astronomer with IBM

Astronomer with IBM, is a data orchestration solution, powered by Apache Airflow®, that brings the simplicity of managing Airflow with enterprise grade functionalities, security and support.



Astronomer with IBM Benefits

Increasing reliability & reducing risk

De-risk data products and project delivery timelines to ensure mission-critical data is delivered reliably and on time.

Improving operational efficiency

Help software and data teams do more with the same or smaller budgets by eliminating undifferentiated heavy-lifting and leveraging organizational network effects.

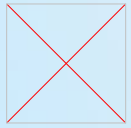
Governance at scale

Enabling organization to apply successful solutions broadly across their organization, supporting multiple teams interacting with data flows and tasks.

Innovating faster

Empower organizations to ship mission-critical data products faster by adopting new technologies quickly.

Govern at the gate, so you can trust on arrival with [watsonx.data integration](#) ↪



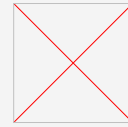
Visit the website for detailed resources, use cases, and insights.

[Go now →](#)



See how it works for you with a free, 30-day trial.

[Try now →](#)



Book a live demo with an IBM technical expert.

[Book now →](#)

